Forces and Moments

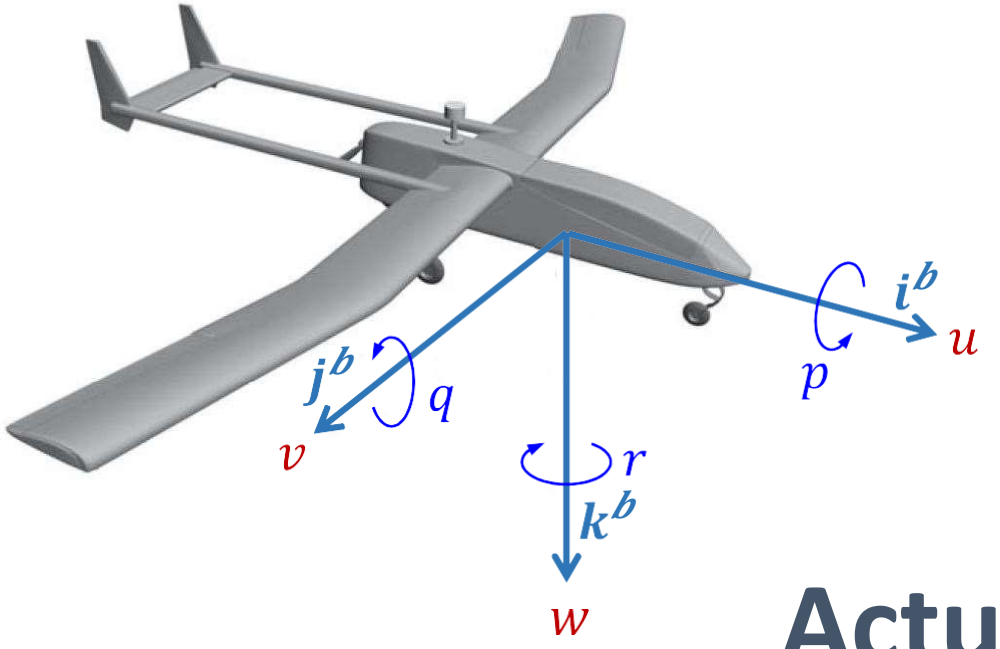
Learning outcomes

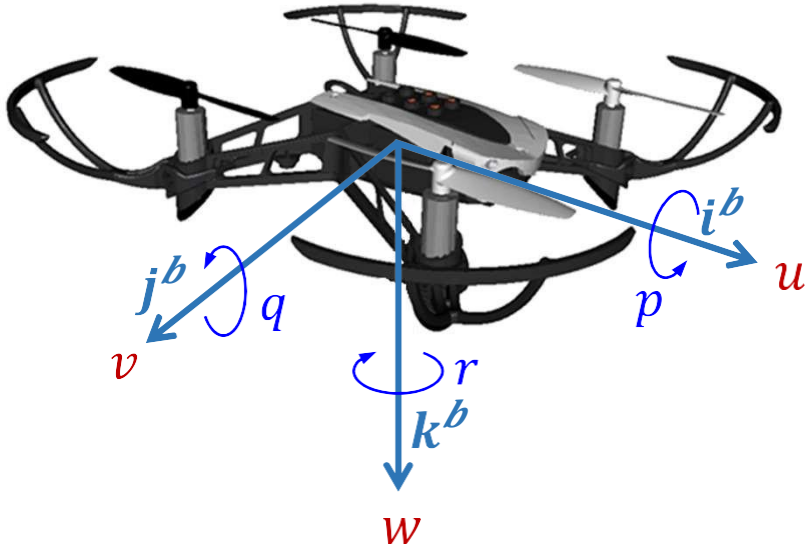
Determine the forces and moments generated by aerial-robot actuators.

Analyse aerodynamic forces and moments.

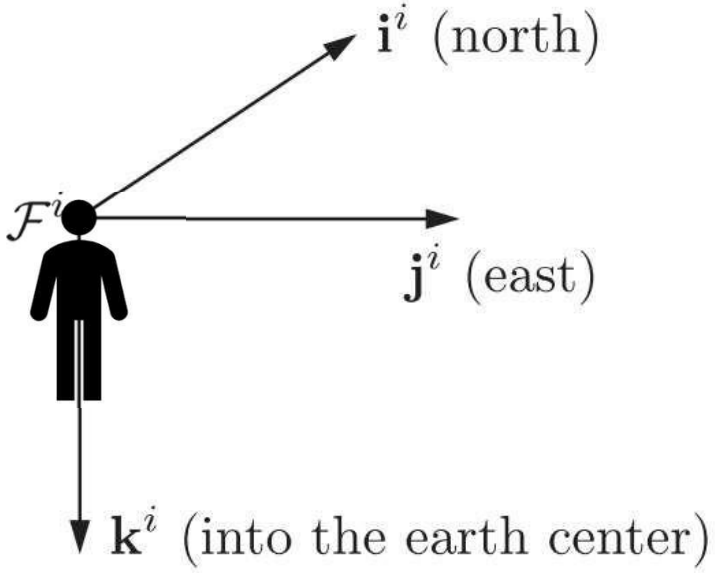
Combine force and moment models with the rigid-body equations of motion.

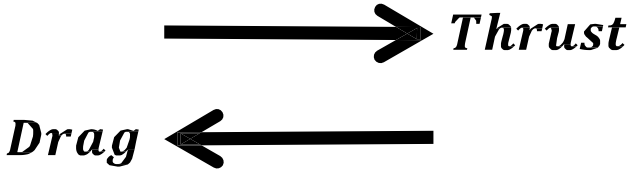
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i^b
u
p
j^b
q
v
r
k^b
w


i^b
u
p
j^b
q
v
r
k^b
w

Actuators


$\mathbf{i}^i$ (north)
$\mathcal{F}^i$
$\mathbf{j}^i$ (east)
$\mathbf{k}^i$ (into the earth center)


Thrust
Drag

Total forces and moments

Gravity, aerodynamics and propulsion contribute to the total force.

$$F^b = F_g^b + F_a^b + F_p^b$$

Moments act about the centre of mass. Gravity has no moment about this point.

$$J\dot{\omega}_b + \omega_b \times (J\omega_b) = M_p^b + M_a^b + M_{gyro}^b$$

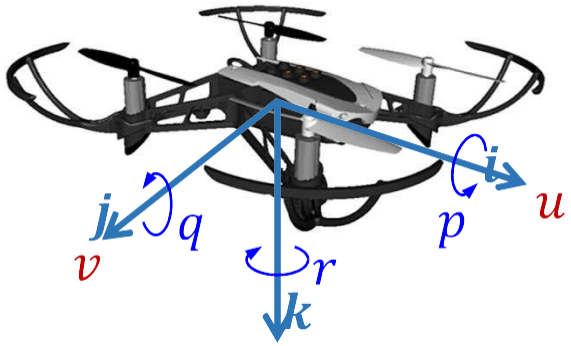
Subscripts: g = gravity, a = aerodynamic, p = propulsion.
The rotor gyroscopic term is defined separately on slides 8–9.

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Gravitational force

In NED, the down axis is positive.

$$F_g^n = \begin{bmatrix} 0 \\ 0 \\ mg \end{bmatrix}$$



In body axes, rotate the same gravity vector:

$$F_g^b = R_n^b F_g^n = mg \begin{bmatrix} -sin\theta \\ sin\varphi cos\theta \\ cos\varphi cos\theta \end{bmatrix}$$

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Aerodynamic drag force

Drag opposes motion relative to the air. Use air-relative velocity, including wind.

$$v_a^b = v_b - v_w^b, V_a = |v_a^b|$$

Low-speed linear approximation

$$F_a^b = -C_{AT} v_a^b$$

Quadratic drag magnitude (direction opposite to the airflow-relative motion)

$$D = \frac{1}{2}\rho V_a^2 A C_D$$

ρ: air density A: reference area
C_D: drag coefficient

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Aerodynamic damping moments

Body angular rates $\omega_b = [p, q, r]^T$ describe rotation about the body axes.

Linear damping approximation

$$M_a^b = -C_{AR}\omega_b$$

One possible quadratic approximation for axis j

$$M_{a,j} = -\frac{1}{2}\rho A L^3 C_{DM,j}|\omega_j|\omega_j$$

L is a reference length. $C_{DM,j}$ is dimensionless.
The sign makes the damping moment oppose the angular rate.

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Rotor gyroscopic moment

Spinning rotors store angular momentum along the body z axis.

$$h_r = He_z, H = J_r(s_1\Omega_1 + \cdots + s_N\Omega_N)$$

J_r: rotor spin inertia. $\Omega_i \geq 0$: speed magnitude.
s_i = +1 for spin about +z, and −1 for spin about −z.

$$M^b_{gyro} = -\omega_b \times h_r = \begin{bmatrix} -qH \\ pH \\ 0 \end{bmatrix}$$

This expression assumes constant rotor spin speeds during the instant considered.

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When can rotor gyroscopic effects be neglected?

$$M_{gyro}^{b} = -\omega_b \times h_r$$

Equal-speed, identical counter-rotating rotors have zero net spin angular momentum.

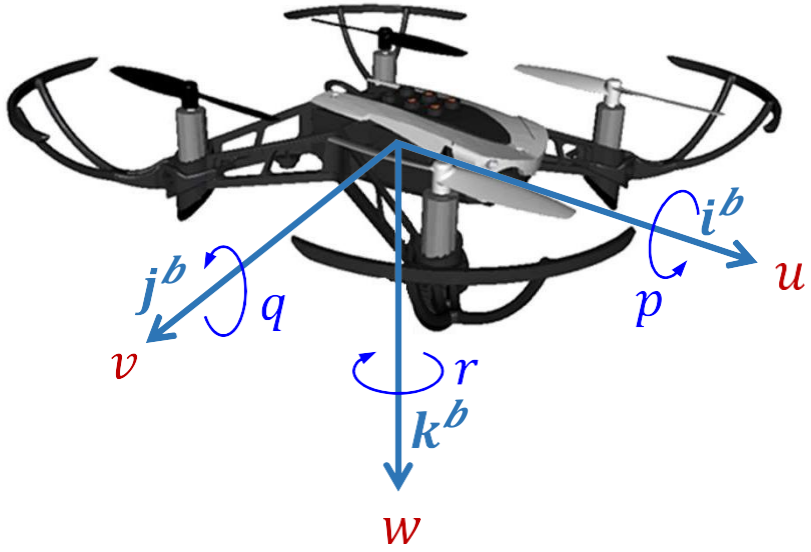
$$H = 0 \Rightarrow M_{gyro}^{b} = 0$$

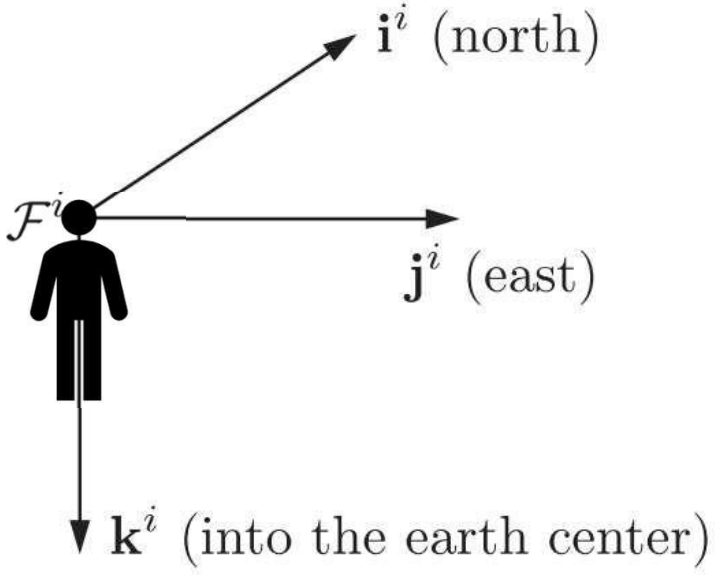
Otherwise, neglect the term only when it is small relative to the other moments.

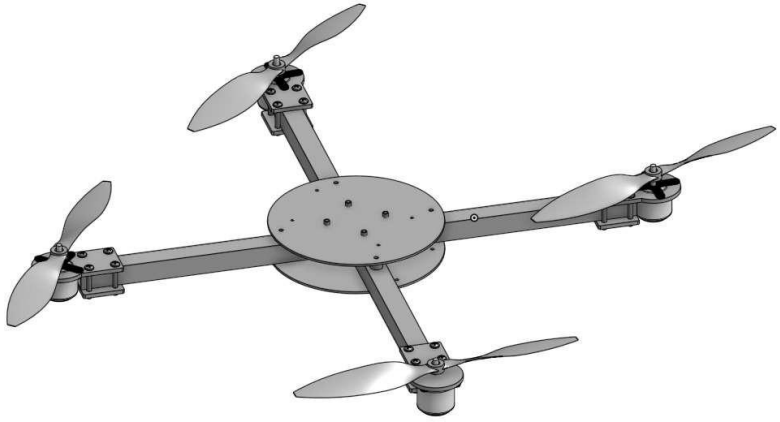
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Multi-Rotor Aerial Robots

Actuator and Aerodynamic Forces





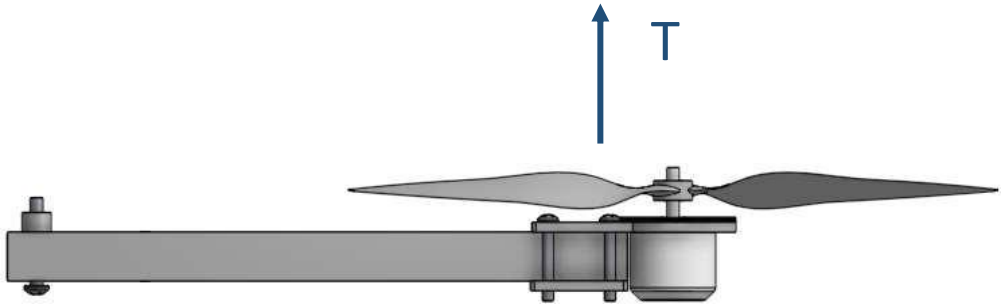


Thrust from a rotating propeller

The resultant thrust acts along the rotor shaft.

$$T = C_T \rho A (\Omega R)^2 = k_T \Omega^2$$

$$A = \pi R^2, k_T = C_T \rho A R^2$$



R: rotor radius Ω: angular speed in rad/s

C_T: dimensionless coefficient using tip speed ΩR

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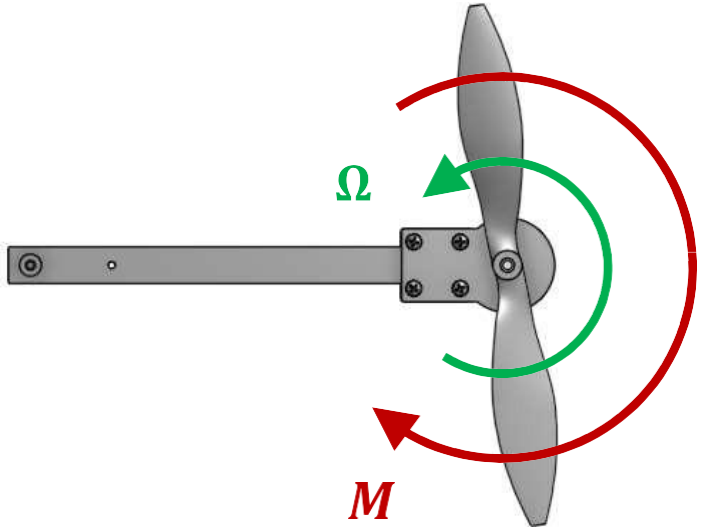
Propeller drag torque

Aerodynamic forces on the blades oppose the rotor's spin.

$$Q = C_M \rho A (\Omega R)^2 R = k_M \Omega^2$$

Q is the torque magnitude.

The motor sustains the rotation.
The airframe experiences an opposite reaction torque.



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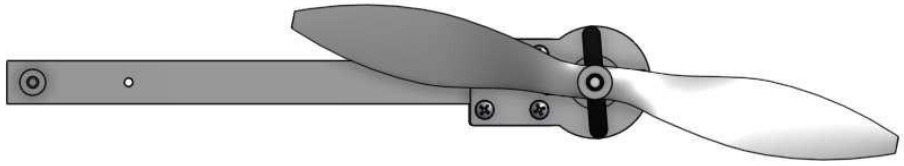
Rotor torque and airframe reaction

Aerodynamic forces on the blades oppose the rotor's spin.

$$Q = C_M \rho A (\Omega R)^2 R = k_M \Omega^2$$

Q is the torque magnitude.

The motor sustains the rotation.
The airframe experiences an opposite reaction torque.



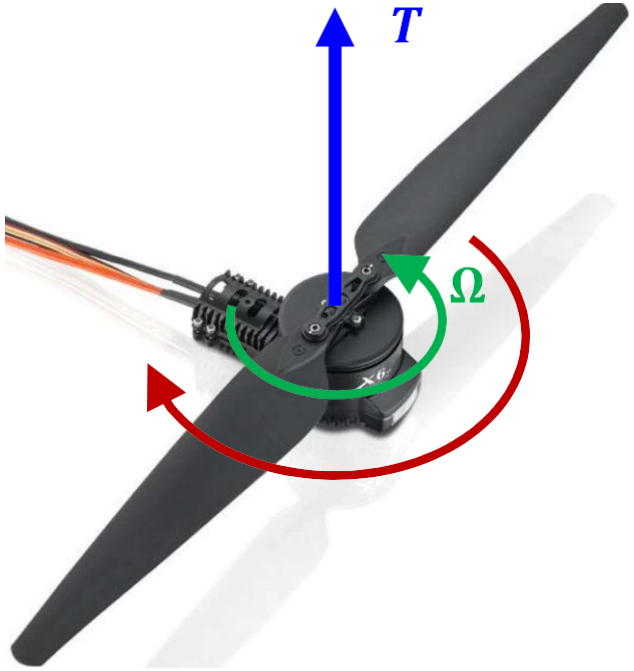
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Rotor thrust and reaction torque

For rotor i

$$T_i = k_T \Omega_i^2$$

$$M_{z,i} = \sigma_i k_M \Omega_i^2$$



σ_i = +1 or −1 sets the reaction-torque direction.

For the numbering used next:
σ = [+1, −1, +1, −1].

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Reaction torque expressed in terms of thrust

$$T_i = k_T \Omega_i^2, M_{z,i} = \sigma_i k_M \Omega_i^2$$

Eliminate the rotor speed:

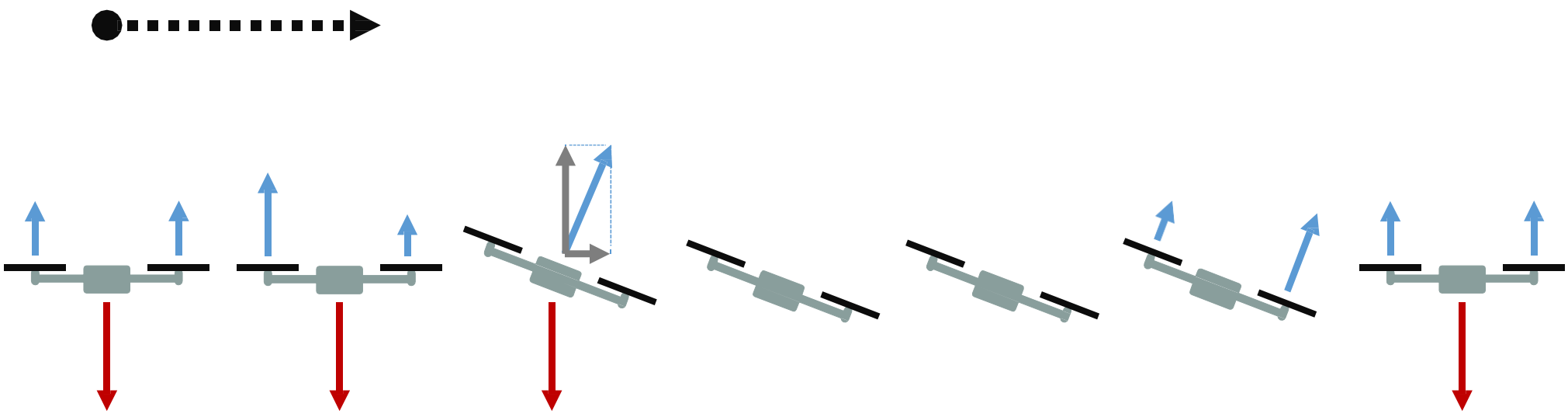
$$M_{z,i} = \sigma_i k_{MT} T_i, k_{MT} = \frac{k_M}{k_T}$$

k_{MT} has units of metres because torque / thrust = N m / N.

Both thrust and torque scale with Ω^2. Therefore, torque is linear in thrust.

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Multi-Rotor Aerial Robots



Tilting redirects thrust. Its horizontal component produces acceleration.

Returning to level removes this horizontal thrust component.
Without drag, existing horizontal velocity does not disappear.

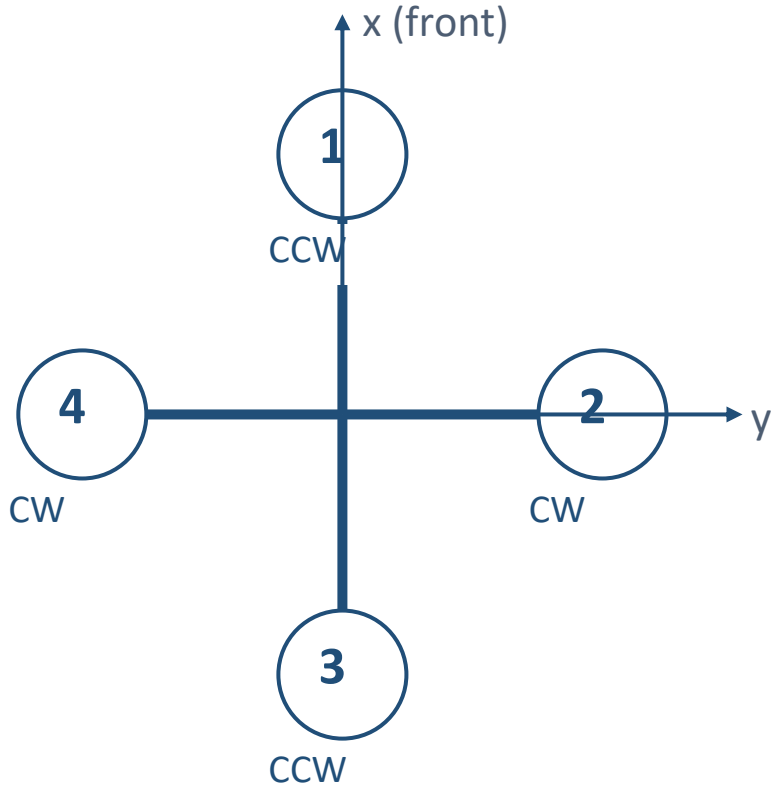
+ configuration: force and moment model

Number the rotors as shown.

Body axes: x forward, y right, z down.

Each thrust T_i acts along −z.

l is the centre-to-rotor distance.



Top view: z into page

$$M_i = r_i \times F_i + M_{z,i} e_z$$

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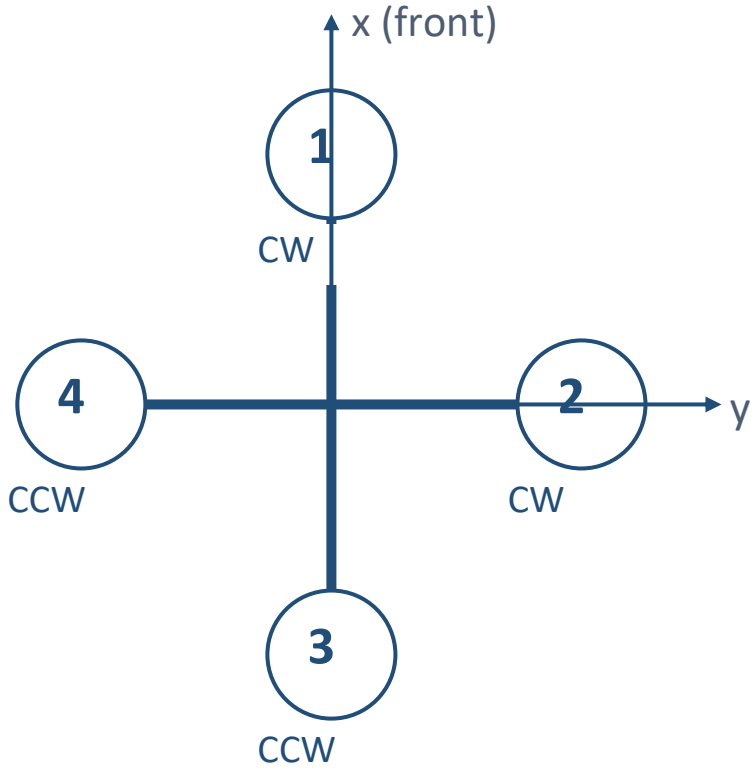
Rotor directions: an unsuitable arrangement

Adjacent rotors with the same spin can cancel yaw torque at equal thrust.

$$M_z = k_{MT}(-T_1 - T_2 + T_3 + T_4)$$

But a pitch command changes the front and rear thrusts in opposite directions.

This also changes yaw torque.
Equal-thrust cancellation is not enough.



Top view: z into page

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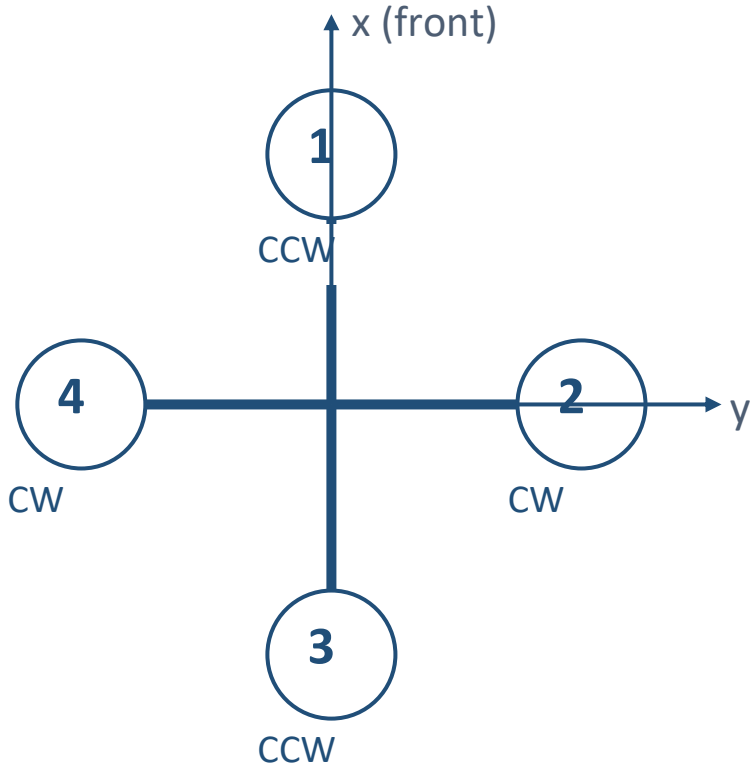
Rotor directions: alternating spin

Opposite rotors have the same spin.
Adjacent rotors have opposite spins.

$$M_z = k_{MT}(T_1 - T_2 + T_3 - T_4)$$

For a pure pitch increment, use equal and opposite changes in T_1 and T_3.

$$\Delta T_1 = \delta, \Delta T_3 = -\delta \Rightarrow \Delta M_z = 0$$



Top view: z into page

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+ configuration: force and moment equations

Rotor positions: 1 front, 2 right, 3 rear, 4 left. Thrust is upward.

$$F_x = F_y = 0, F_z = -(T_1 + T_2 + T_3 + T_4)$$

$$M_x = l(T_4 - T_2)$$

$$M_y = l(T_1 - T_3)$$

$$M_z = k_{MT}(T_1 - T_2 + T_3 - T_4)$$

These are propulsive force and moment components. Add gravity and aerodynamic loads separately.

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+ configuration: motor speeds

$$\begin{bmatrix} F_z \\ M_x \\ M_y \\ M_z \end{bmatrix} = \begin{bmatrix} -1 & -1 & -1 & -1 \\ 0 & -l & 0 & l \\ l & 0 & -l & 0 \\ k_{MT} & -k_{MT} & k_{MT} & -k_{MT} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} = A_+ t$$

$$t = A_+^{-1} \begin{bmatrix} F_z \\ M_x \\ M_y \\ M_z \end{bmatrix} \quad , \Omega_i = \sqrt{\frac{T_i}{k_T}}$$

A feasible command requires $0 \leq T_i \leq T_{max}$ for every rotor.

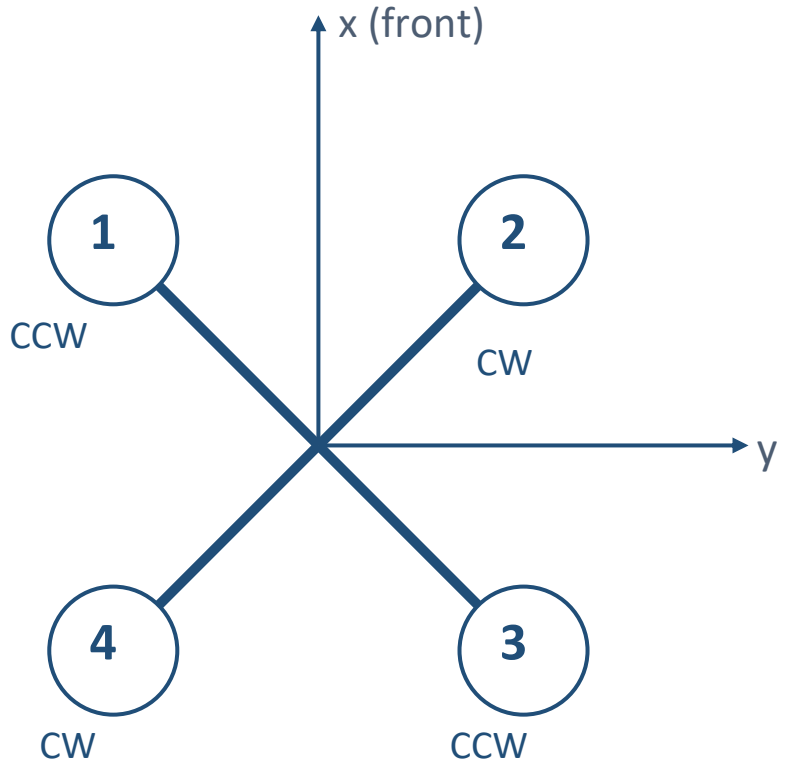
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× configuration: rotor numbering

1: front-left 2: front-right
3: rear-right 4: rear-left

l is the radial arm length.
Each x and y offset has magnitude d.

$$d = \frac{l}{\sqrt{2}}$$



Top view: z into page

Choose alternating rotor spins around the aircraft.

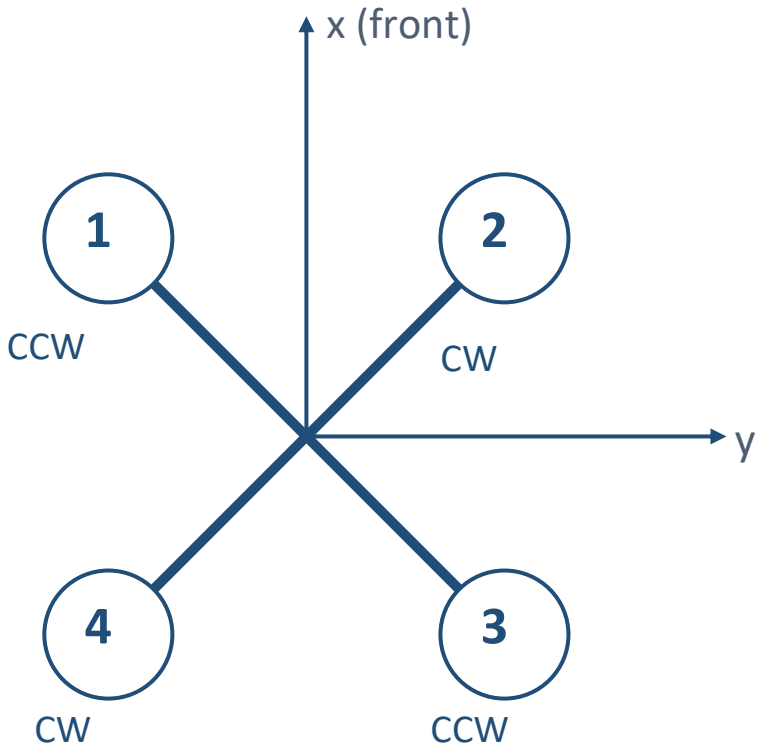
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× configuration: why alternate the spins?

Front pair spinning alike:
a pitch command can also create yaw.

Left pair spinning alike:
a roll command can also create yaw.

Alternating spins:
roll and pitch increments can cancel their yaw-torque increments.



Top view: z into page

Keep the numbering and spin convention consistent in the allocation matrix.

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× configuration: force and moment equations

$$d = \frac{l}{\sqrt{2}}, F_z = -(T_1 + T_2 + T_3 + T_4)$$

$$M_x = d(T_1 - T_2 - T_3 + T_4)$$

$$M_y = d(T_1 + T_2 - T_3 - T_4)$$

$$M_z = k_{MT}(T_1 - T_2 + T_3 - T_4)$$

Use $r_i \times F_i$ for each rotor. $F_x = F_y = 0$ in body axes.

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× configuration: motor speeds

$$\begin{bmatrix} F_z \\ M_x \\ M_y \\ M_z \end{bmatrix} = \begin{bmatrix} -1 & -1 & -1 & -1 \\ d & -d & -d & d \\ d & d & -d & -d \\ k_{MT} & -k_{MT} & k_{MT} & -k_{MT} \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} = A_{\times} t$$

$$t = A_{\times}^{-1} \begin{bmatrix} F_z \\ M_x \\ M_y \\ M_z \end{bmatrix} \quad , \Omega_i = \sqrt{\frac{T_i}{k_T}}$$

d = l/√2. Check the thrust limits after allocating the command.

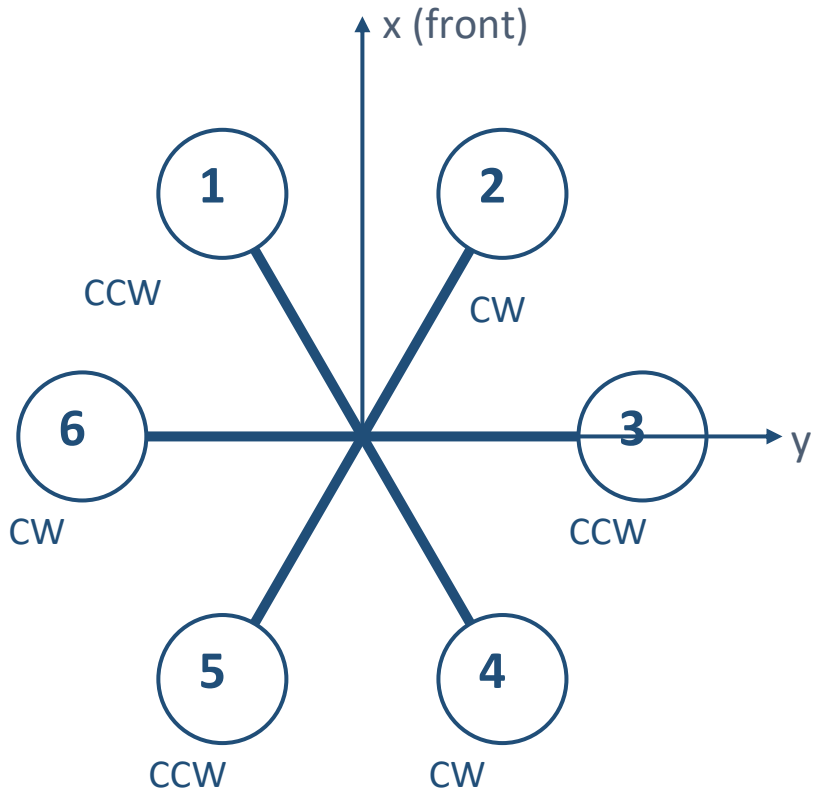
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Hexarotor: geometry and rotor numbering

Six equal arms of length l.
Body x points between rotors 1 and 2.

Measure β_i from +x towards +y.

$$r_i = \begin{bmatrix} l\cos\beta_i \\ l\sin\beta_i \\ 0 \end{bmatrix}$$



Top view: z into page

β = [−30°, 30°, 90°, 150°, 210°, 270°]
Reaction signs alternate: +, −, +, −, +, −.

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Hexarotor: force and moment allocation

For each rotor, the allocation column is $[-1, -y_i, x_i, \sigma_i k_{MT}]^T$.

$$\begin{bmatrix} F_z \\ M_x \\ M_y \\ M_z \end{bmatrix} = \begin{bmatrix} -1 & -1 & -1 & -1 & -1 & -1 \\ a & -a & -l & -a & a & l \\ b & b & 0 & -b & -b & 0 \\ k & -k & k & -k & k & -k \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \\ T_5 \\ T_6 \end{bmatrix}$$

$$a = \frac{l}{2}, b = \frac{\sqrt{3}l}{2}, k = k_{MT}$$

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Hexarotor: motor speeds

Six thrusts produce four commanded components.
The allocation is not unique.

For this regular geometry, one unconstrained minimum-norm allocation is

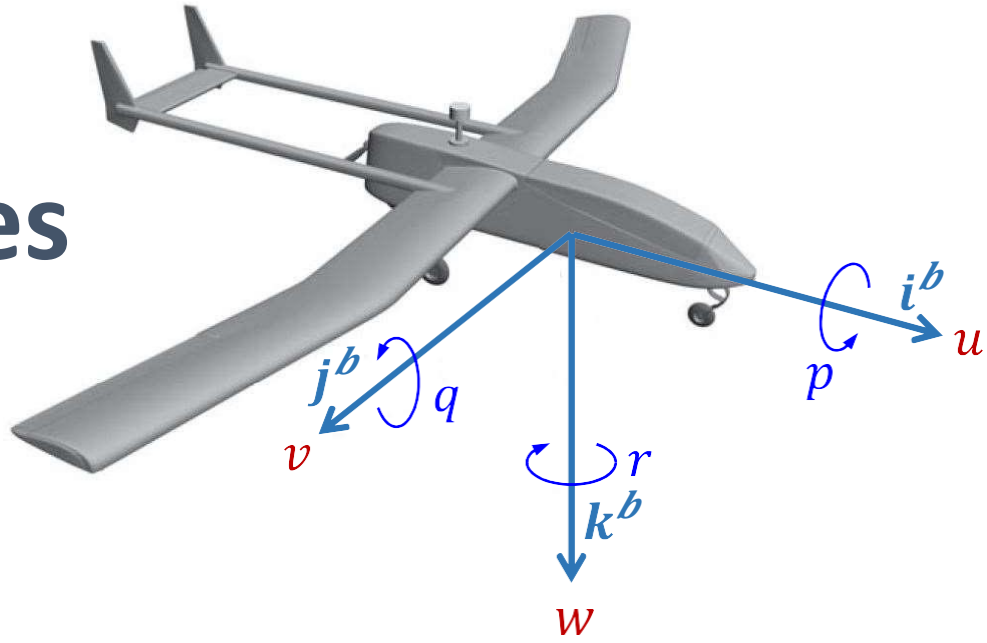
$$T_i = -\frac{F_z}{6} - \frac{y_i M_x}{3l^2} + \frac{x_i M_y}{3l^2} + \frac{\sigma_i M_z}{6k_{MT}}$$

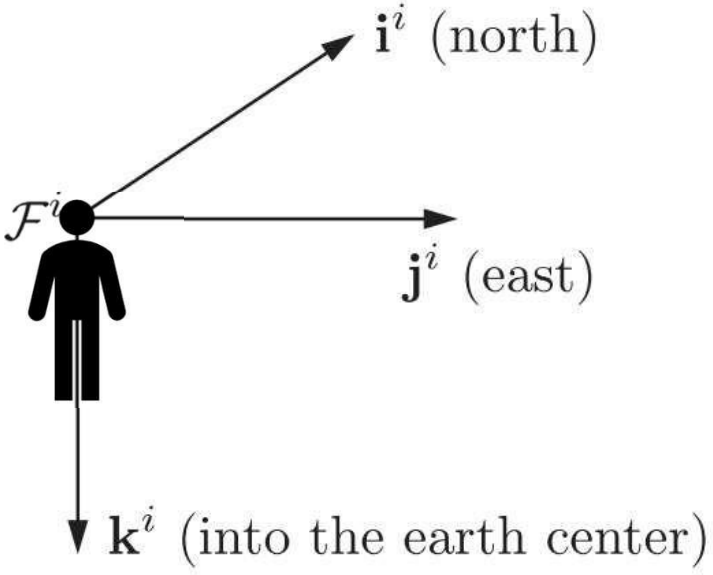
$$\Omega_i = \sqrt{\frac{T_i}{k_T}}$$

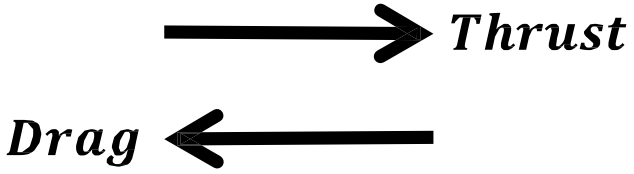
Check all thrust limits. If a limit is violated, this unconstrained solution is not feasible.

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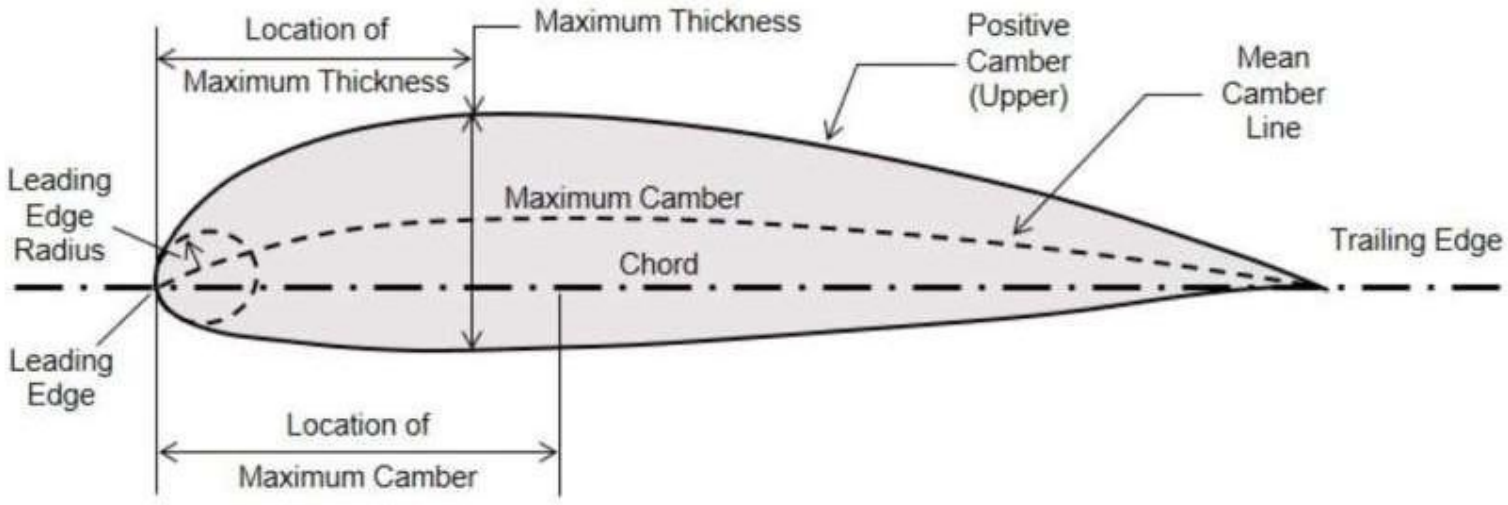
Aerodynamic Forces and Moments (Airfoil Theory)







Airfoil terminology

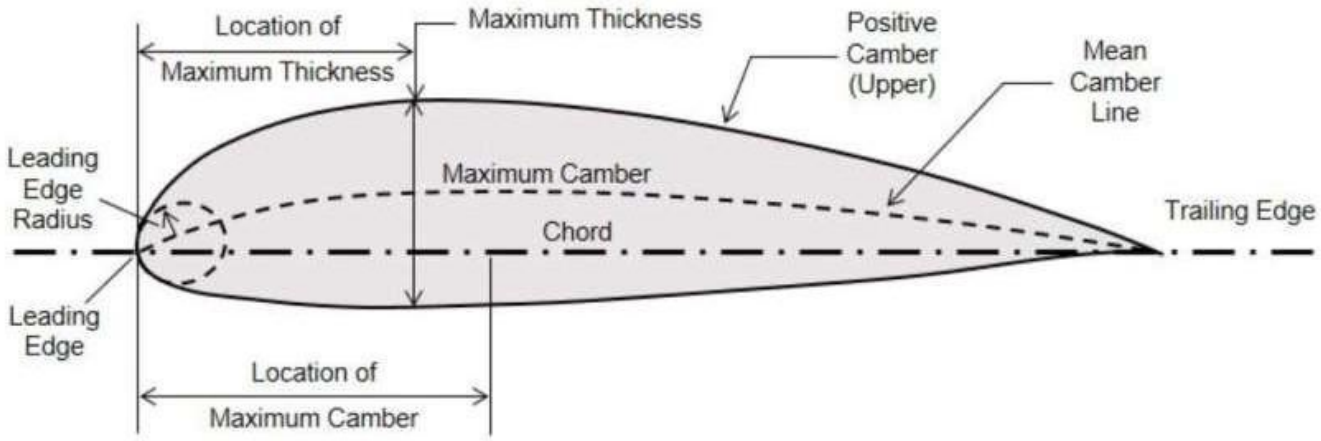


Mean camber line: points halfway between the upper and lower surfaces.
Leading / trailing edges: front and rear ends of the airfoil.
Chord line: straight line joining the leading and trailing edges.
Chord, c: distance between those edges along the chord line.
Camber: maximum separation of camber line and chord line,
measured perpendicular to the chord line.

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NACA four-digit airfoil designation

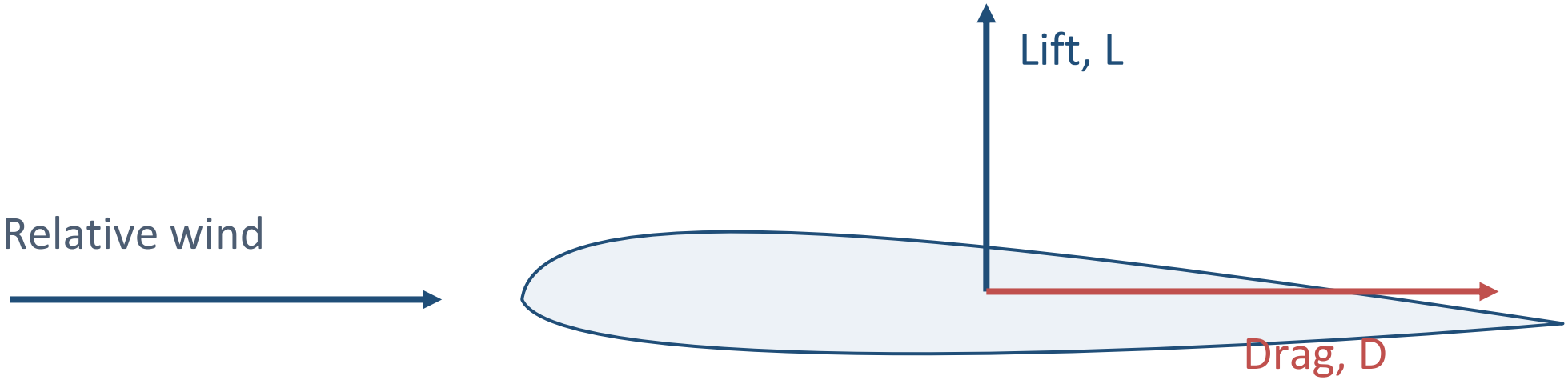
First digit: maximum camber as a percentage of chord.
Second digit: camber location in tenths of chord.
Last two digits: maximum thickness as a percentage of chord.



NACA 2415: 2% camber at 40% chord, with 15% maximum thickness.

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Airflow, lift and drag

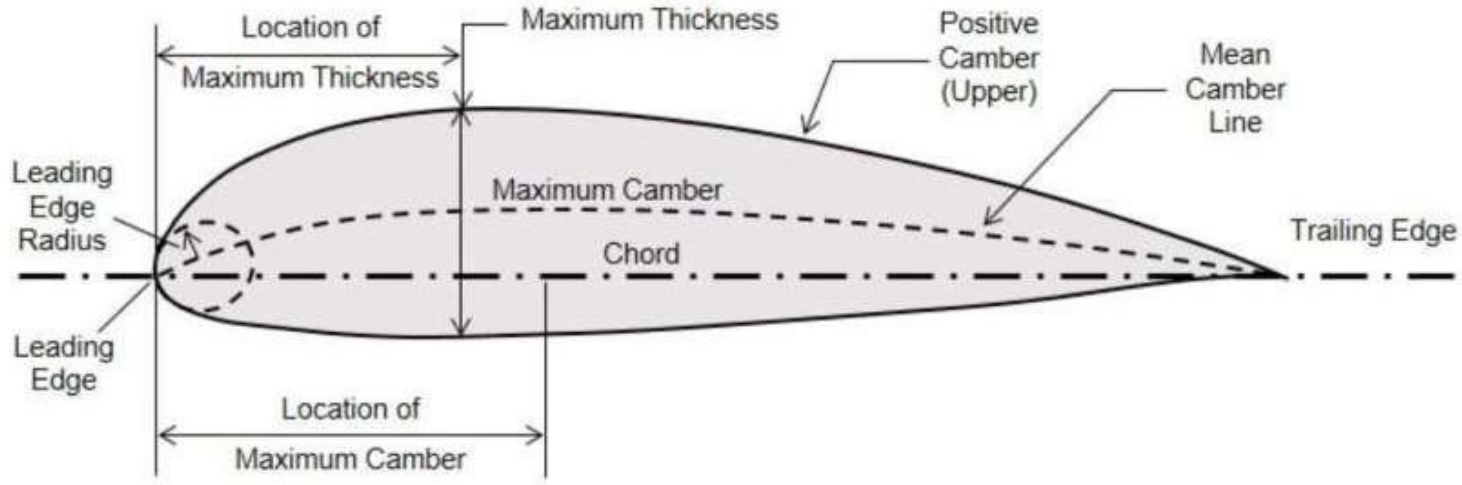


Lift is perpendicular to the relative wind.
Drag acts in the direction of the relative wind.

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Airfoil section and aircraft axes

An airfoil is the cross-sectional shape of a wing or rotor blade.



The chord line belongs to the airfoil section.
The body x axis belongs to the aircraft.
They can differ by a fixed mounting angle, called wing incidence.

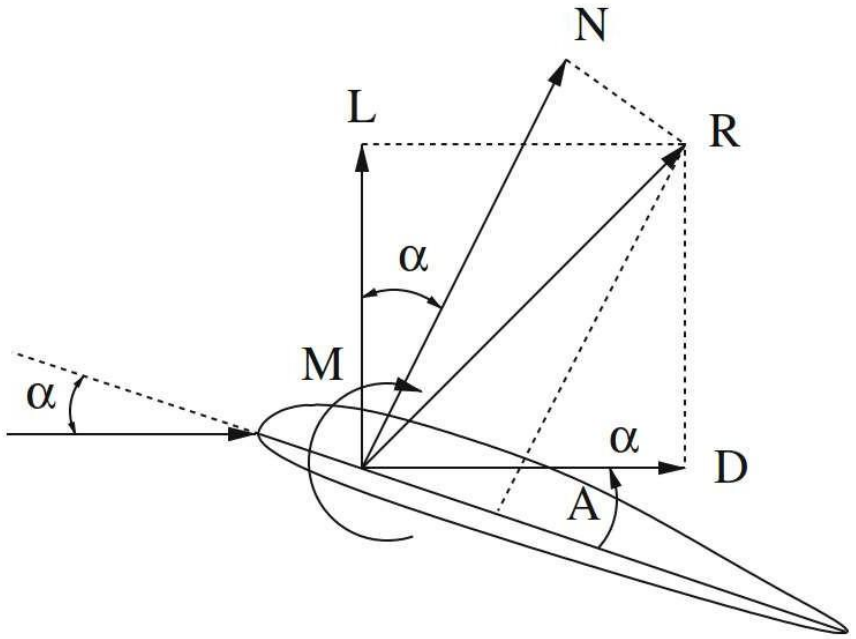
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Aerodynamic quantities

Relative wind: air velocity relative to the aircraft.
The subscript ∞ means far upstream.

Angle of attack, α: angle between the chord and the oncoming-flow line.

Lift: perpendicular to relative wind.
Drag: parallel to relative wind.

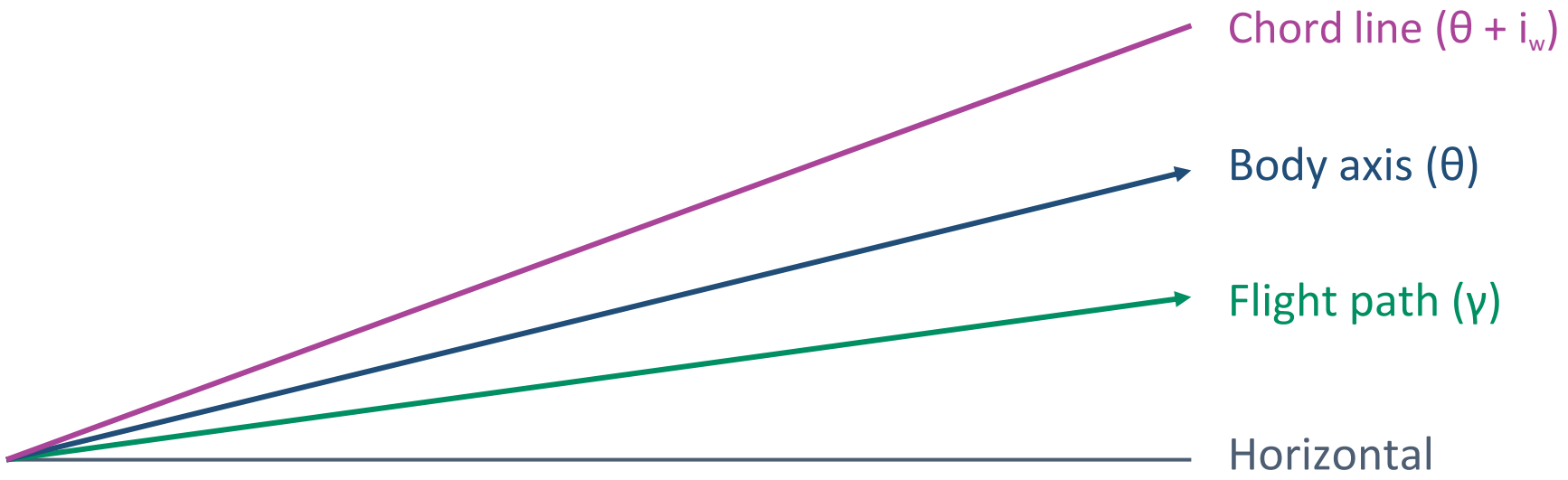


The pitching moment depends on its reference point.
The pressure centre has zero moment from the resultant pressure loading.

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Angle of attack and wing incidence

Wing incidence is fixed by the mounting. Angle of attack changes with the flight condition.

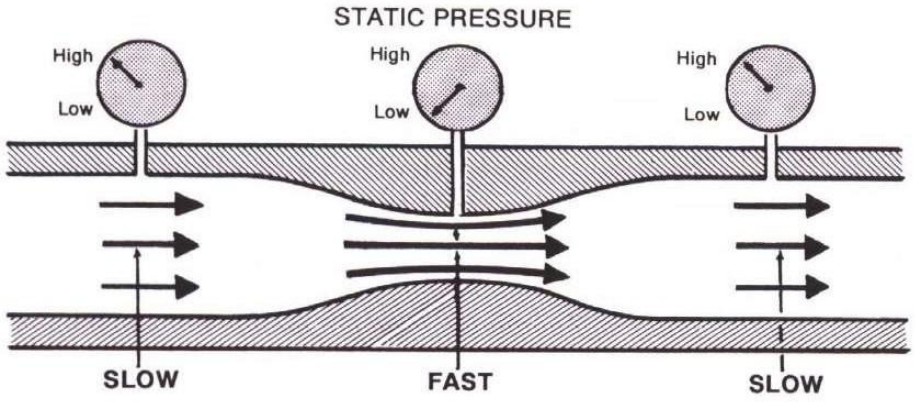


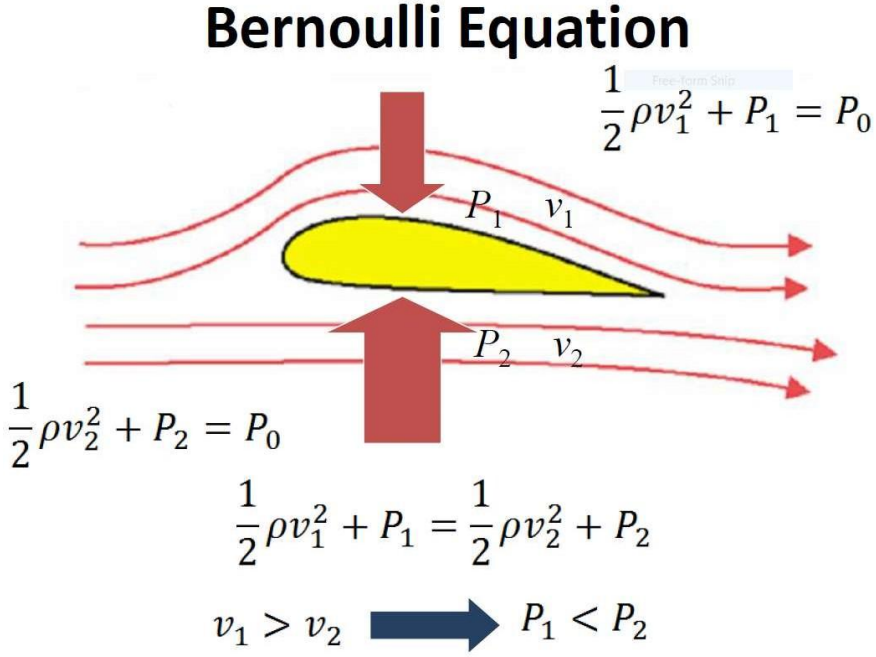
$$\alpha = \theta + i_w - \gamma$$

α: angle of attack i_w: wing incidence γ: air-relative flight-path angle

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Bernoulli's equation and pressure





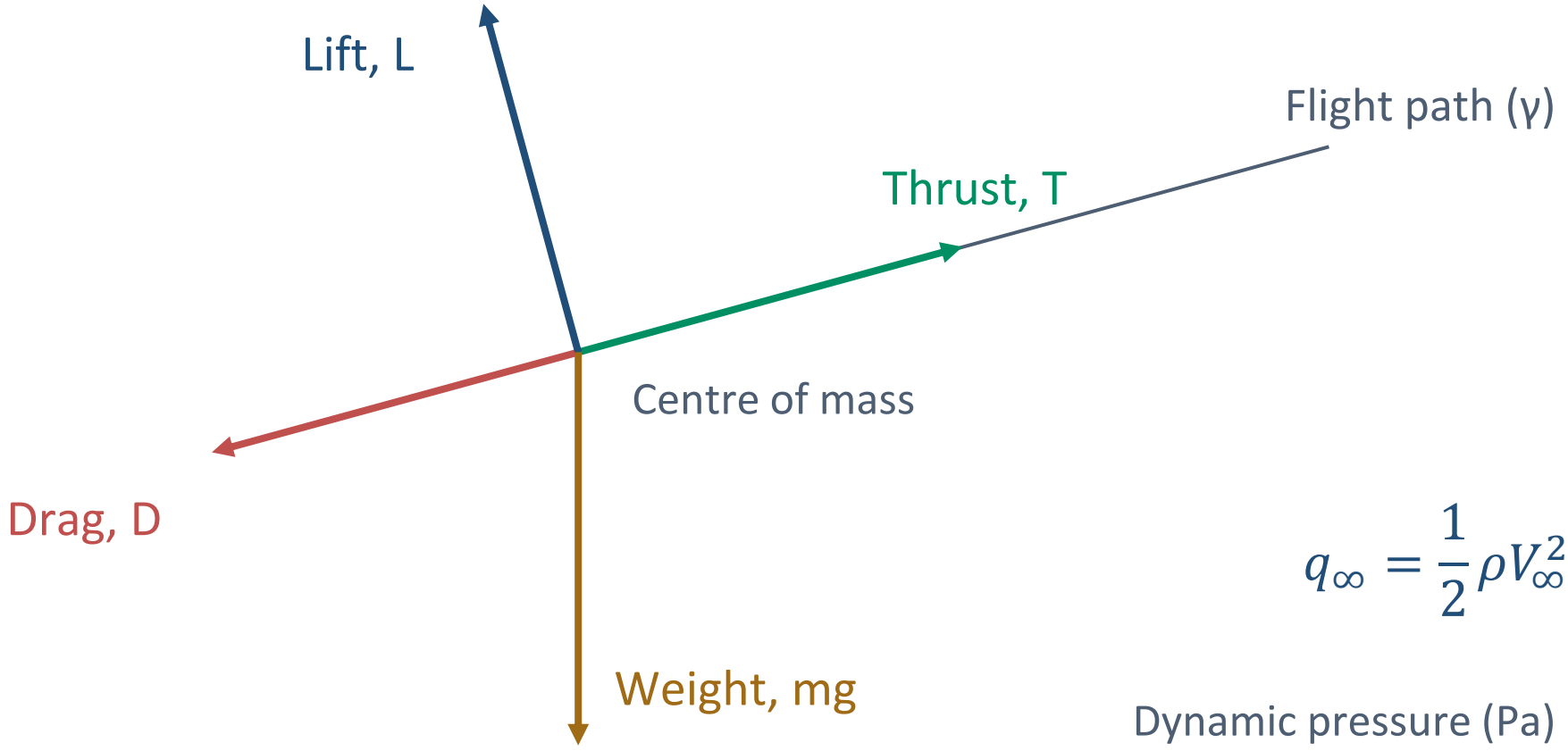
$$p + \frac{1}{2}\rho V^2 + \rho g h = constant$$

Steady, incompressible, inviscid flow along a streamline.
Pressure differences over the surface contribute to lift.

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Forces on a fixed-wing aircraft

Side view: steady climb with thrust parallel to the flight path



$$q_\infty = \frac{1}{2}\rho V_\infty^2$$

Dynamic pressure (Pa)

$$L = mg\cos\gamma, T - D = mg\sin\gamma$$

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Lift, drag and moment coefficients

$$q_\infty = \frac{1}{2}\rho V_\infty^2$$

$$L = q_\infty S C_L, D = q_\infty S C_D$$

$$M = q_\infty S c C_m$$

$$\boxed{M = \frac{1}{2}\rho V_\infty^2 S c c_m}$$

$$c_m \equiv \frac{M}{\frac{1}{2}\rho V_\infty^2 S c} = \frac{M}{q_\infty S c}$$

$$c_m = f_3(\alpha, \mathrm{Re})$$

S: reference wing area
c: reference chord
C_L, C_D, C_m: dimensionless coefficients

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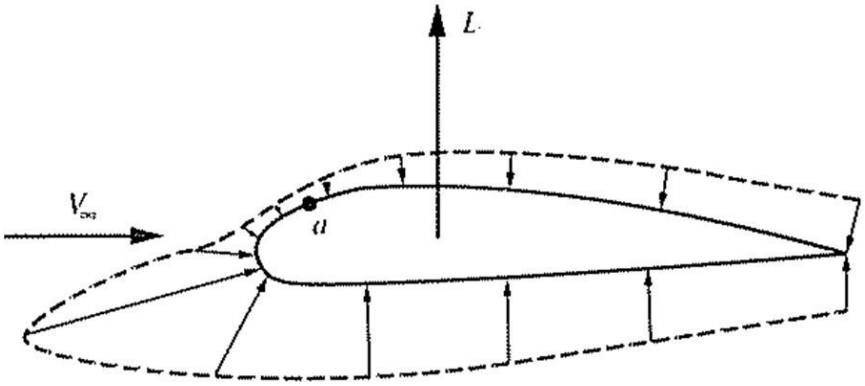
Pressure coefficient

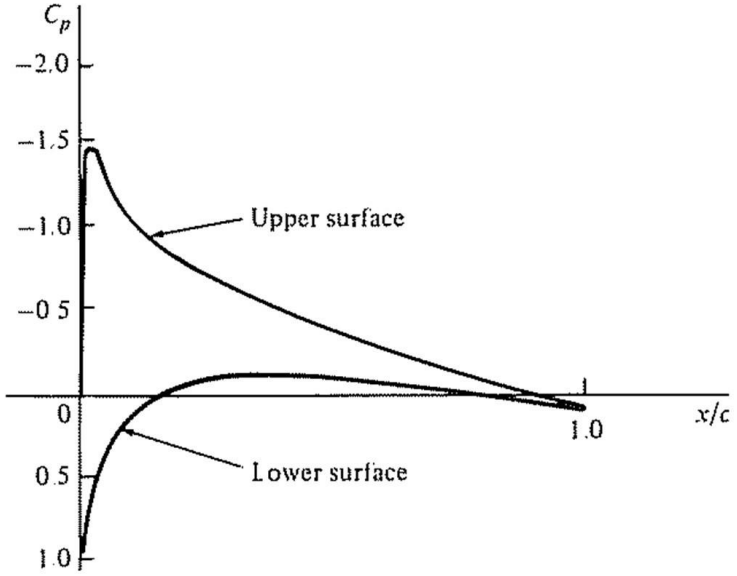
A dimensionless description of surface pressure

$$C_p = \frac{p - p_\infty}{\frac{1}{2}\rho V_\infty^2} = \frac{p - p_\infty}{q_\infty}$$

p: local static pressure

p∞: upstream static pressure

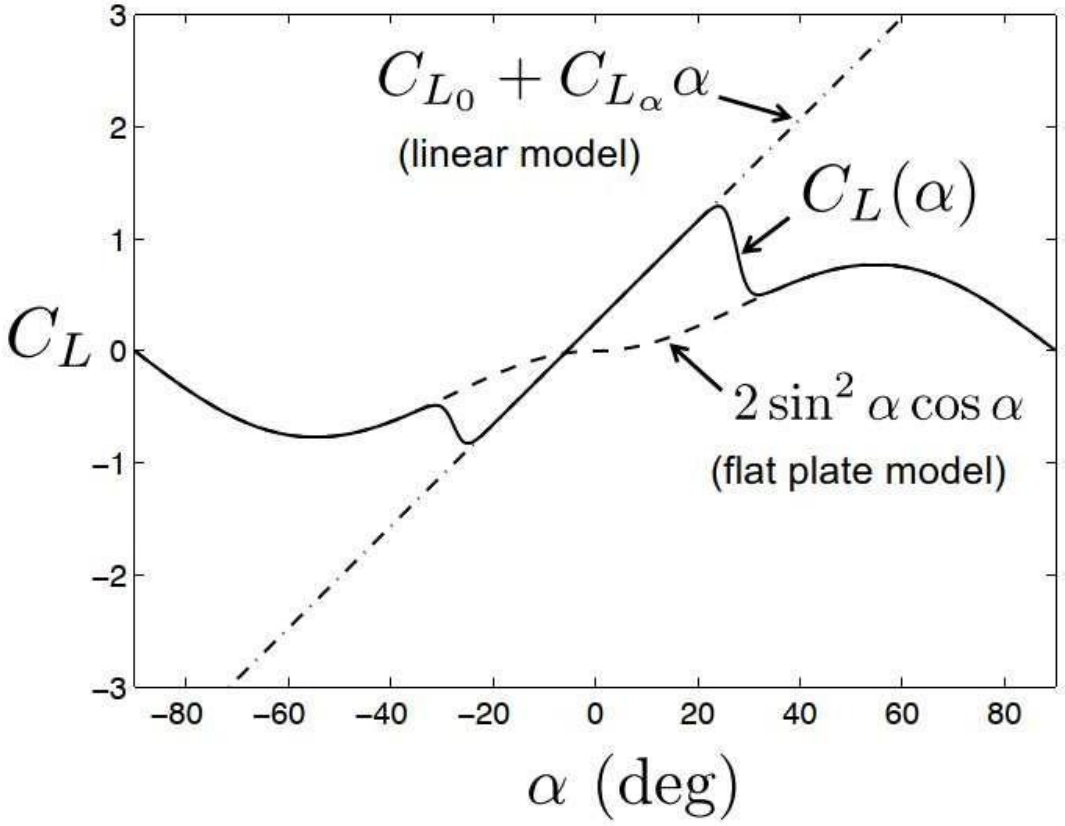


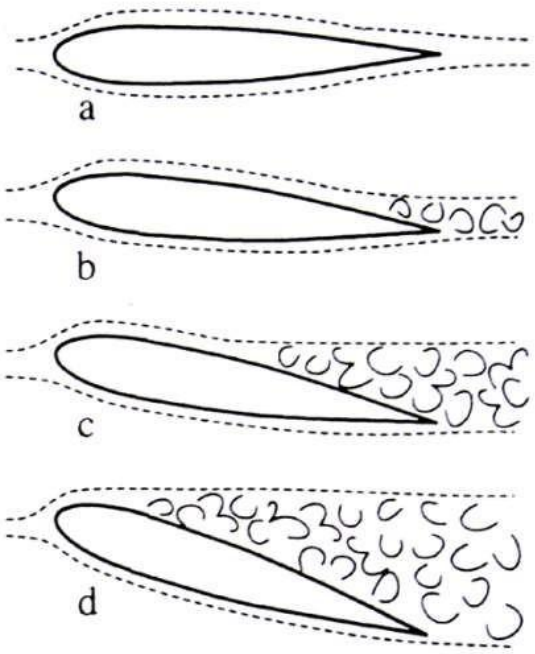


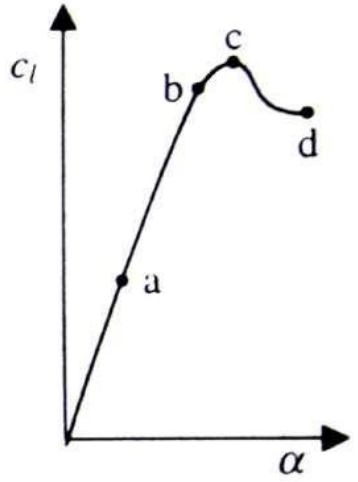
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Lift coefficient and angle of attack

Lift varies approximately linearly with α before stall.





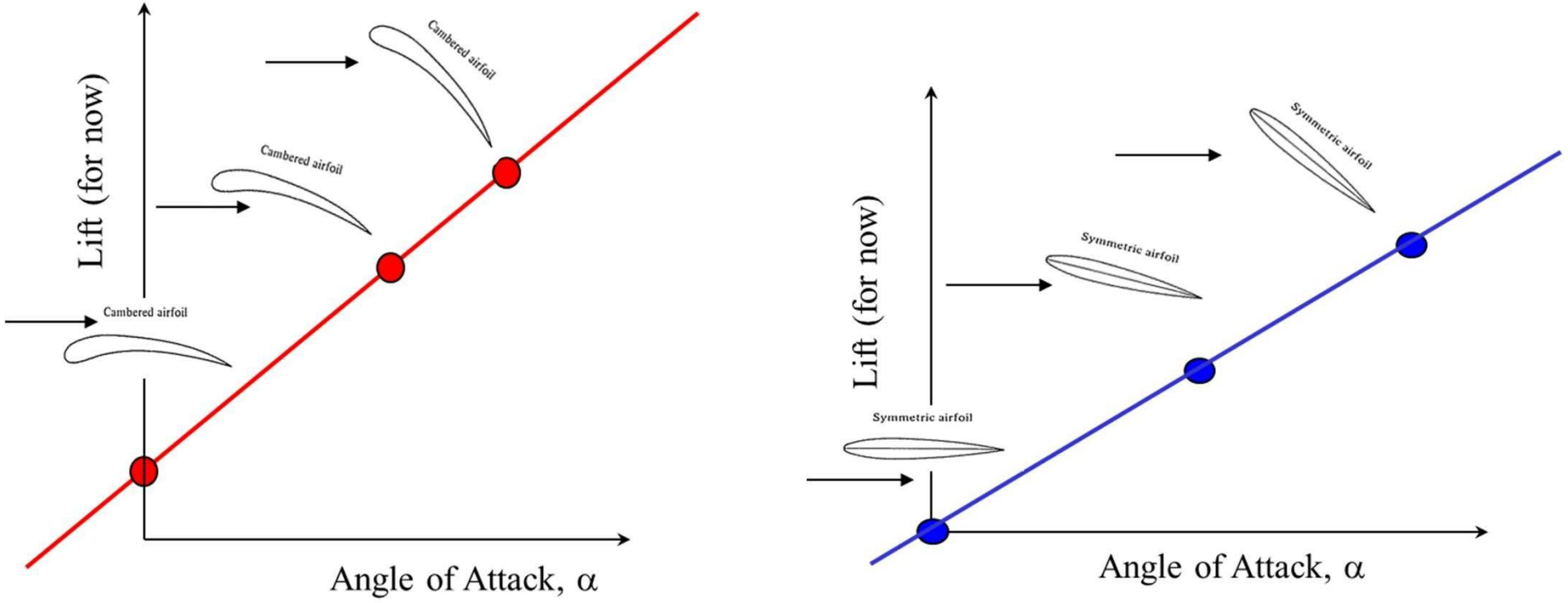


$$L = q_\infty S C_L, D = q_\infty S C_D, M = q_\infty S c C_m$$

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Cambered and symmetric airfoils

Camber shifts the zero-lift angle.
A symmetric airfoil has approximately zero lift at α = 0.

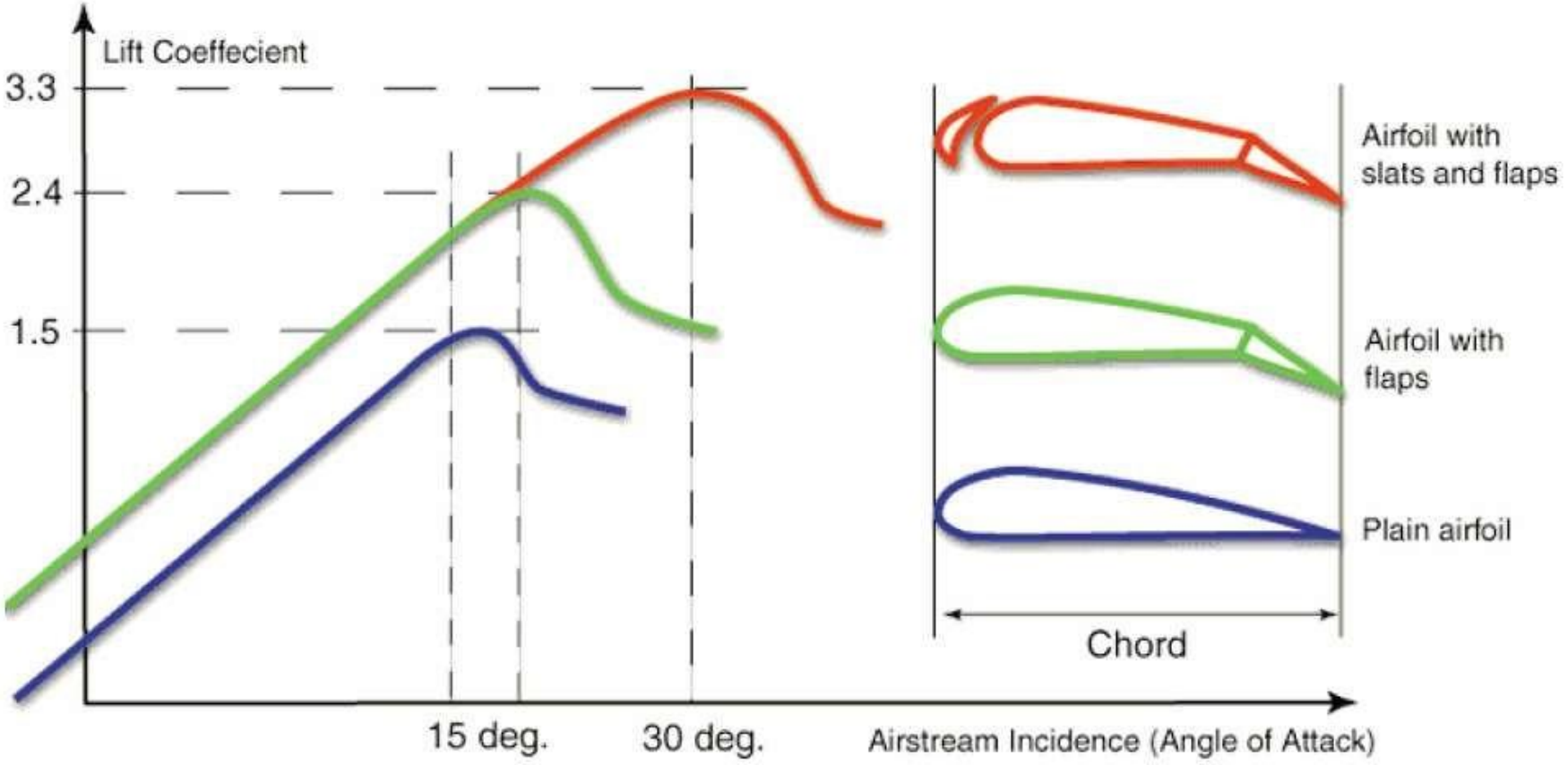


$$L = q_\infty S C_L, D = q_\infty S C_D, M = q_\infty S c C_m$$

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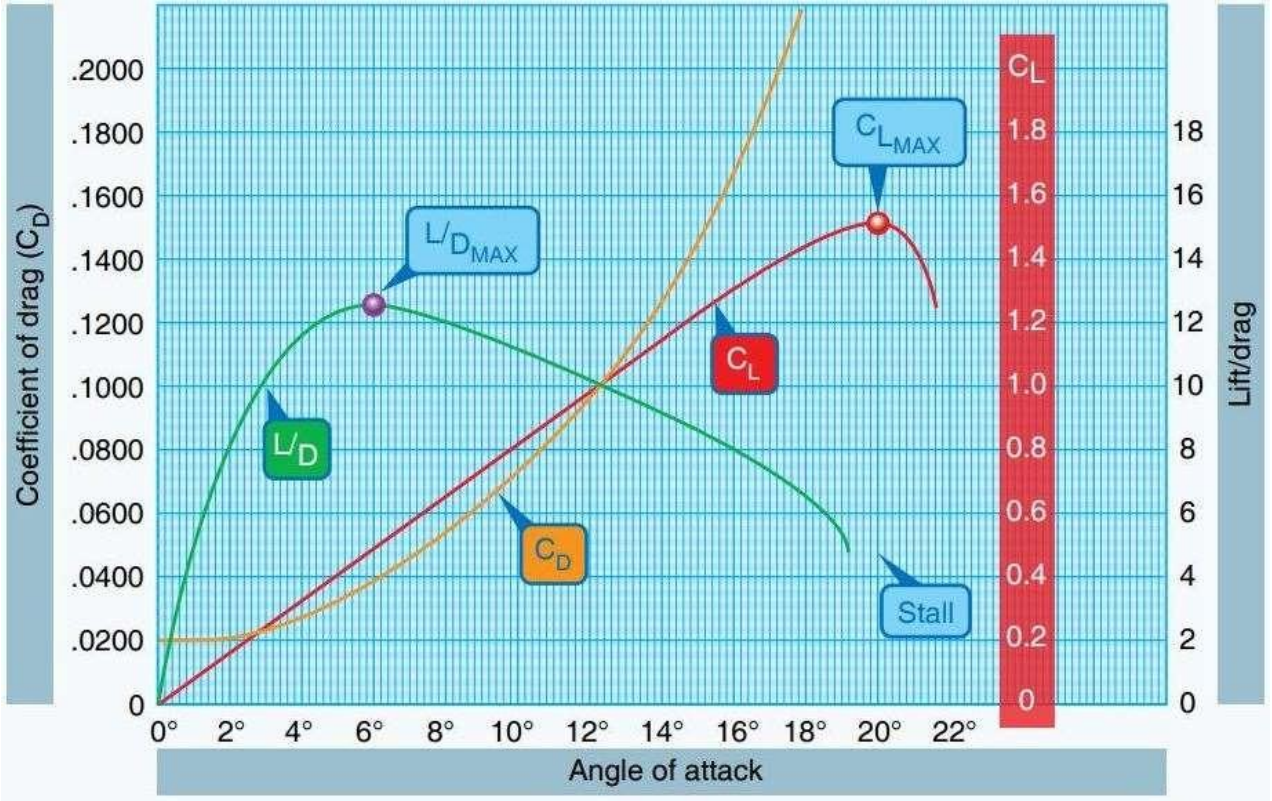
Flaps and slats

High-lift devices change the lift curve and maximum lift coefficient.



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Lift-to-drag ratio



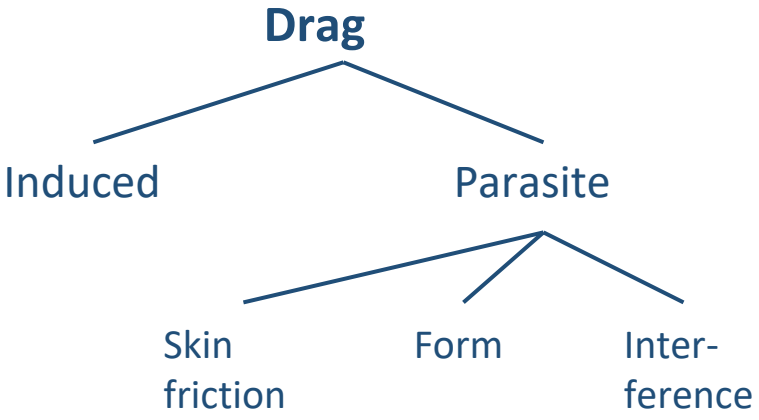
$$\frac{L}{D} = \frac{C_L}{C_D}$$

Maximum L/D occurs at a particular flight condition, not necessarily at maximum lift.

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Types of aerodynamic drag

Induced drag accompanies lift generation by a finite wing.
At fixed speed and geometry, it grows approximately with lift squared.



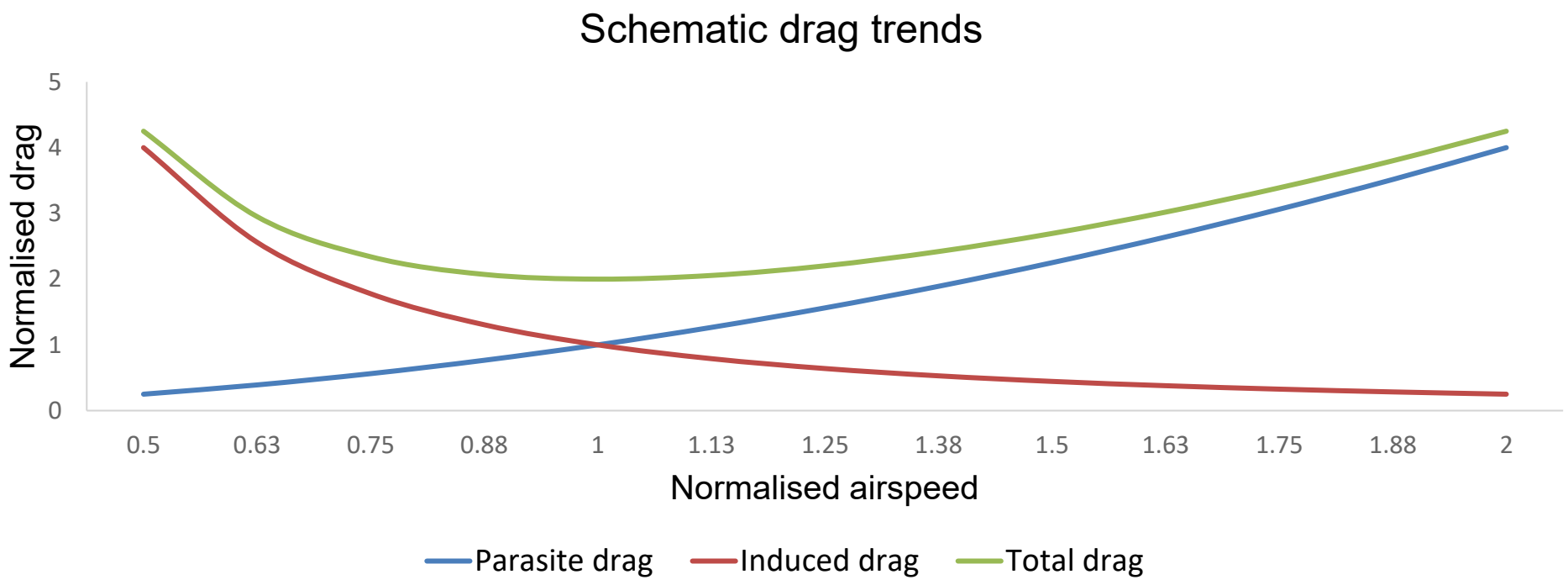
Parasite drag includes:
Form drag: pressure effects associated with shape.
Skin-friction drag: viscous shear over the surface.
Interference drag: interaction between components.

Airfoil profile drag generally combines skin-friction and pressure drag.
Do not count it again as a separate addition to the same components.

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Drag versus airspeed

Steady level flight: hold lift equal to weight.



Parasite drag $\propto V^2$. Induced drag $\propto 1/V^2$ at constant lift.

The coefficients and scales here are illustrative, not aircraft test data.

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Summary: forces and moments in the model

Multirotor propulsion: rotor speeds determine thrust and reaction torque.

$$T_i = k_T \Omega_i^2, M_{z,i} = \sigma_i k_{MT} T_i$$

Fixed-wing aerodynamics: airspeed, attitude and geometry determine the aerodynamic loads.

$$L = q_\infty S C_L, D = q_\infty S C_D, M = q_\infty S c C_m$$

Insert the signed total forces and moments into the rigid-body equations.

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Appendices

Appendix: rigid-body equations and quaternion form

Constant mass and body inertia. All force and moment components use body axes.

$$\dot{p}_n = R_b^n v_b, \dot{v}_b = \frac{F^b}{m} - \omega_b \times v_b$$

$$\dot{\omega}_b = J^{-1}(M^b - \omega_b \times (J\omega_b))$$

Unit quaternion η: scalar first, Hamilton product, body-to-NED rotation.

$$\dot{\eta} = \frac{1}{2} \begin{bmatrix} 0 & -p & -q & -r \\ p & 0 & r & -q \\ q & -r & 0 & p \\ r & q & -p & 0 \end{bmatrix} \eta, |\eta| = 1$$

F^b includes gravity. M^b includes the retained external and rotor-coupling terms.

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Appendix: diagonal inertia and Euler angles

cα = cos α, sα = sin α. R maps body velocity into NED.

$$R_b^n = \begin{bmatrix} c_\theta c_\psi & s_\varphi s_\theta c_\psi - c_\varphi s_\psi & c_\varphi s_\theta c_\psi + s_\varphi s_\psi \\ c_\theta s_\psi & s_\varphi s_\theta s_\psi + c_\varphi c_\psi & c_\varphi s_\theta s_\psi - s_\varphi c_\psi \\ -s_\theta & s_\varphi c_\theta & c_\varphi c_\theta \end{bmatrix}$$

$$\dot{\varphi} = p + sin\varphi tan\theta\ q + cos\varphi tan\theta\ r$$

$$\dot{\theta} = cos\varphi\ q - sin\varphi\ r, \dot{\psi} = \frac{sin\varphi\ q + cos\varphi\ r}{cos\theta}$$

$$\dot{p} = \frac{M_x + (J_y - J_z)qr}{J_x}, \dot{q} = \frac{M_y + (J_z - J_x)pr}{J_y}$$

$$\dot{r} = \frac{M_z + (J_x - J_y)pq}{J_z}, J = diag(J_x, J_y, J_z)$$

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Appendix: nonzero product of inertia

$$J = \begin{bmatrix} J_x & 0 & -J_{xz} \\ 0 & J_y & 0 \\ -J_{xz} & 0 & J_z \end{bmatrix}, J^{-1} = \begin{bmatrix} J_z/\Gamma & 0 & J_{xz}/\Gamma \\ 0 & 1/J_y & 0 \\ J_{xz}/\Gamma & 0 & J_x/\Gamma \end{bmatrix}$$

$$\dot{p} = \Gamma_1 pq - \Gamma_2 qr + \Gamma_3 M_x + \Gamma_4 M_z$$

$$\dot{q} = \Gamma_5 pr - \Gamma_6 (p^2 - r^2) + M_y/J_y$$

$$\dot{r} = \Gamma_7 pq - \Gamma_1 qr + \Gamma_4 M_x + \Gamma_8 M_z$$

$$\Gamma = J_x J_z - J_{xz}^2, \Gamma_1 = \frac{J_{xz}(J_x - J_y + J_z)}{\Gamma}, \Gamma_2 = \frac{J_z(J_z - J_y) + J_{xz}^2}{\Gamma}$$

$$\Gamma_3 = \frac{J_z}{\Gamma}, \Gamma_4 = \frac{J_{xz}}{\Gamma}, \Gamma_5 = \frac{J_z - J_x}{J_y}, \Gamma_6 = \frac{J_{xz}}{J_y}$$

$$\Gamma_7 = \frac{J_x(J_x - J_y) + J_{xz}^2}{\Gamma}, \Gamma_8 = \frac{J_x}{\Gamma}$$

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